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USE doc_hate;

-- 1. articles_with_authors_and_orgs
-- Purpose: This view combines information from articles, authors, and
organizations
-- to provide a comprehensive overview of each article with its authors and
publishing organization.
DROP VIEW IF EXISTS articles_with_authors_and_orgs;
CREATE VIEW articles_with_authors_and_orgs AS
SELECT a.Article_ID, a.Article_Title, a.Article_Date,
       GROUP_CONCAT(DISTINCT au.Author_Name SEPARATOR ', ') AS Authors,
       o.Organization_Name
FROM articles a
JOIN article_author aa ON a.Article_ID = aa.Article_ID
JOIN author au ON aa.Author_ID = au.Author_ID
JOIN organizations o ON a.Organization_ID = o.Organization_ID
GROUP BY a.Article_ID, a.Article_Title, a.Article_Date, o.Organization_Name;

-- 2. articles_with_keywords
-- Purpose: This view shows articles published between June and August 2017 along
with their associated keywords,
-- allowing for analysis of trending topics during this specific time period.
DROP VIEW IF EXISTS articles_with_keywords;
CREATE VIEW articles_with_keywords AS
SELECT a.Article_ID, a.Article_Title, a.Article_Date,
       GROUP_CONCAT(DISTINCT k.Keyword SEPARATOR ', ') AS Keywords
FROM articles a
JOIN articles_keywords ak ON a.Article_ID = ak.Article_ID
JOIN keywords k ON ak.Keyword_ID = k.Keyword_ID
WHERE a.Article_Date BETWEEN '2017-06-01' AND '2017-08-31'
GROUP BY a.Article_ID, a.Article_Title, a.Article_Date;

-- 3. articles_per_state
-- Purpose: This view provides a count of articles for each state, helping to
identify
-- which states have the most coverage or incidents related to hate crimes.
DROP VIEW IF EXISTS articles_per_state;
CREATE VIEW articles_per_state AS
SELECT s.State_Name, COUNT(a.Article_ID) AS Article_Count
FROM articles a
JOIN locations l ON a.Location_ID = l.Location_ID
JOIN states s ON l.State_ID = s.State_ID
GROUP BY s.State_Name
ORDER BY Article_Count DESC;

-- 4. organization_article_stats
-- Purpose: This view provides statistics on articles published by each
organization,
-- including total articles, average article length, and their most recent article
date.
DROP VIEW IF EXISTS organization_article_stats;
CREATE VIEW organization_article_stats AS
SELECT
    o.Organization_ID,
    o.Organization_Name,
    COUNT(a.Article_ID) AS Total_Articles,
    AVG(LENGTH(a.Summary)) AS Avg_Summary_Length,
    MAX(a.Article_Date) AS Most_Recent_Article_Date,
    (SELECT COUNT(DISTINCT ak.Keyword_ID)
     FROM articles_keywords ak
     WHERE ak.Article_ID = a.Article_ID) AS Total_Keywords
FROM organizations o
JOIN articles a ON o.Organization_ID = a.Organization_ID

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        FROM articles_keywords ak
        WHERE ak.Article_ID IN (SELECT Article_ID FROM articles WHERE Organization_ID
= o.Organization_ID)
        ) AS Total_Unique_Keywords
FROM
    organizations o
LEFT JOIN
    articles a ON o.Organization_ID = a.Organization_ID
GROUP BY
    o.Organization_ID, o.Organization_Name
HAVING
    Total_Articles > (SELECT AVG(article_count) FROM (SELECT COUNT(*) as
article_count FROM articles GROUP BY Organization_ID) AS avg_counts)
ORDER BY
    Total_Articles DESC;

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-- 5. hate_crime_articles
-- Purpose: This view focuses specifically on articles related to hate crimes,
-- helping to track and analyze coverage of these incidents over time.
DROP VIEW IF EXISTS hate_crime_articles;
CREATE VIEW hate_crime_articles AS
SELECT a.Article_ID, a.Article_Title, a.Article_Date, o.Organization_Name
FROM articles a
JOIN organizations o ON a.Organization_ID = o.Organization_ID
WHERE a.Article_ID IN (
    SELECT ak.Article_ID
    FROM articles_keywords ak
    JOIN keywords k ON ak.Keyword_ID = k.Keyword_ID
    WHERE k.Keyword LIKE '%hate%' OR k.Keyword LIKE '%crime%' OR k.Keyword LIKE
'%violence%'
)
ORDER BY a.Article_Date DESC;

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-- Test 1: articles_with_authors_and_orgs
SELECT * FROM articles_with_authors_and_orgs LIMIT 5;

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-- Test 2: articles_with_keywords
SELECT * FROM articles_with_keywords LIMIT 5;

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-- Test 3: articles_per_state
SELECT * FROM articles_per_state;

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-- Test 4: organization_article_stats
SELECT * FROM organization_article_stats;

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-- Test 5: hate_crime_articles
SELECT * FROM hate_crime_articles LIMIT 5;

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-- Additional interesting queries:
-- 6. Monthly Trend of Hate Crime Reporting
SELECT
    DATE_FORMAT(a.Article_Date, '%Y-%m') AS Month,
    COUNT(DISTINCT a.Article_ID) AS Article_Count
FROM
    articles a
JOIN
    articles_keywords ak ON a.Article_ID = ak.Article_ID
JOIN

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        keywords k ON ak.Keyword_ID = k.Keyword_ID
WHERE
    k.Keyword LIKE '%hate%' OR k.Keyword LIKE '%crime%'
GROUP BY
    Month
ORDER BY
    Month;

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-- 7. Geographic Distribution of Hate Crime Reporting

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SELECT
    s.State_Name,
    COUNT(DISTINCT a.Article_ID) AS Total_Articles,
    COUNT(DISTINCT CASE WHEN k.Keyword LIKE '%hate%' OR k.Keyword LIKE '%crime%'
THEN a.Article_ID END) AS Hate_Crime_Articles,
    ROUND(COUNT(DISTINCT CASE WHEN k.Keyword LIKE '%hate%' OR k.Keyword LIKE
'%crime%' THEN a.Article_ID END) * 100.0 / COUNT(DISTINCT a.Article_ID), 2) AS
Percentage_Hate_Crime_Articles
FROM
    states s
JOIN
    locations l ON s.State_ID = l.State_ID
JOIN
    articles a ON l.Location_ID = a.Location_ID
LEFT JOIN
    articles_keywords ak ON a.Article_ID = ak.Article_ID
LEFT JOIN
    keywords k ON ak.Keyword_ID = k.Keyword_ID
GROUP BY
    s.State_Name
ORDER BY
    Hate_Crime_Articles DESC;

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